CA-005

In 2008 an individual under name Satoshi Nakamoto introduced Bitcoin aiming to introduce a decentralized digital currency that allows direct online payments b/w parties without relying on financial institutions.

- Decentralized transactions: Bitcoin enables P2P transactions removing need for intermediaries like banks
- Double Spending Solution: To prevent this issue Bitcoin uses a public ledger ~~like~~ Blockchain, where transactions are recorded and verified by net participants

- Proof of Work Mechanism: Transaction are grouped into blocks. Miners compete to solve Mathematical puzzles first to solve adds his block to the chain, earning a reward. This process secures the network and maintains consensus.

- Network Consensus: As long as majority of computing power is controlled by honest nodes, the system remains secure. Nodes accept longest chain as valid one ensuring agreement on transaction history.

- Privacy and Security: Transactions are public, however user identities remain anonymous. Security is maintained through cryptographic techniques, making it impractical to alter past transactions.